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Industrial Internship Report on

"URL SHORTNER"

Prepared by

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Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. We had to finish the project including the report in 6 weeks' time. My project was (My assigned project was **"Python-Based URL Shortener System."** The objective of this project is to develop an application that converts long and complex web URLs into short, unique, and easily shareable links. The system stores the relationship between the original URL and the shortened URL in a database. Whenever a user opens the shortened link, the application automatically redirects them to the original website. The project is designed using Python and aims to demonstrate concepts such as backend development, URL validation, database management, and HTTP redirection.) Throughout the internship, I enhanced my understanding of Python programming, problem-solving, debugging, database connectivity, and application design. This internship provided valuable industry exposure and improved both my technical knowledge and professional skills, preparing me for future software development projects and career opportunities. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

TABLE OF CONTENTS

1 Preface

Industrial internships play an important role in bridging the gap between academic learning and practical industry experience. They allow students to understand how theoretical concepts are applied in real-world projects while developing technical and professional skills. This internship, organized by **Upskill Campus** in collaboration with **The IoT Academy** and **UniConverge Technologies Pvt. Ltd. (UCT)**, provided an opportunity to work on an industry-inspired Python project and gain hands-on experience in software development.

The assigned project, **URL Shortener System**, focuses on developing an application that converts lengthy URLs into shorter, unique links for easier sharing and management. Such systems are widely used in digital platforms, marketing campaigns, social media, and business applications where concise and manageable links improve usability and user experience.

During this internship, I explored Python programming concepts, backend application development, database management, debugging techniques, and software testing methodologies. Working on this project helped me strengthen my programming fundamentals, analytical thinking, and problem-solving abilities. I also gained an understanding of project planning, system design, and the importance of writing clean, efficient, and maintainable code.

I sincerely thank **Upskill Campus**, **The IoT Academy**, **UniConverge Technologies Pvt. Ltd.**, my mentors, trainers, and **Amity University Jharkhand** for providing this valuable opportunity and continuous guidance throughout the internship. Their support has significantly contributed to my technical growth and career development.

I encourage my fellow students to actively participate in industrial internships, as they provide valuable exposure to professional software development practices and prepare students for future career opportunities in the IT industry.

2 Introduction

The rapid growth of the internet has made sharing web resources a common activity in both personal and professional environments. However, many websites generate long and complex URLs that are difficult to remember, share, or include in emails, social media posts, and documents. A URL Shortener solves this problem by converting long web addresses into short, unique, and user-friendly links while preserving the original destination.

During this internship, I was assigned the project **"Python-Based URL Shortener System."** The project aims to design and develop a system capable of generating shortened URLs and redirecting users to the original web pages. The application demonstrates the practical use of Python programming, backend development, database management, and web technologies.

The internship provided an opportunity to understand how software applications are planned, designed, developed, and tested in an industrial environment. Besides technical knowledge, the internship also enhanced my problem-solving ability, analytical thinking, teamwork, and understanding of professional software development practices.

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

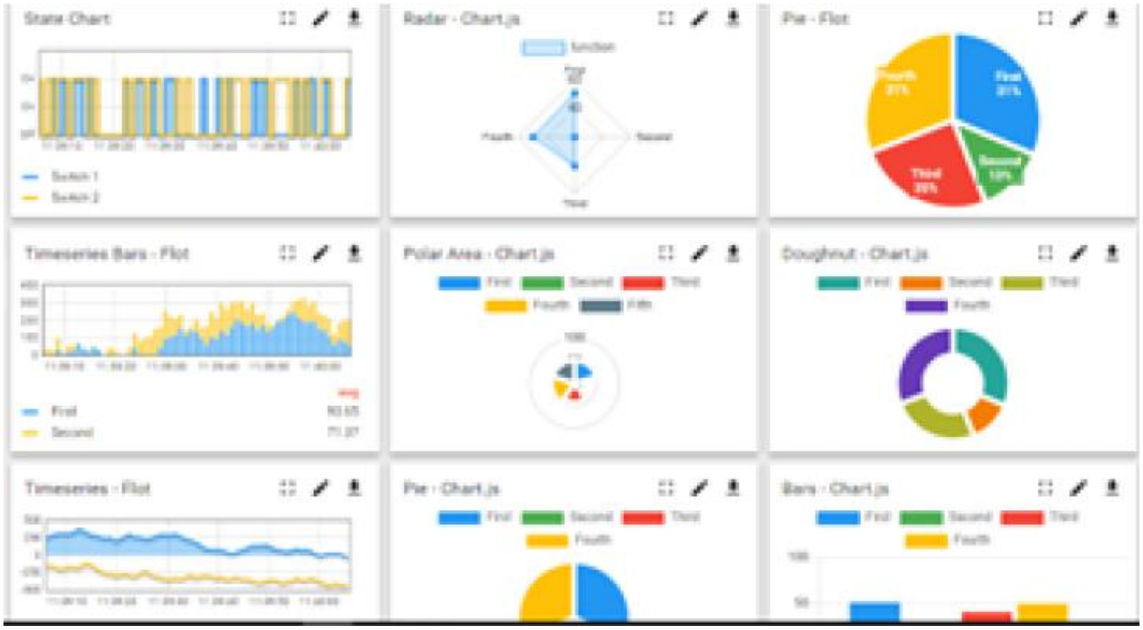
For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

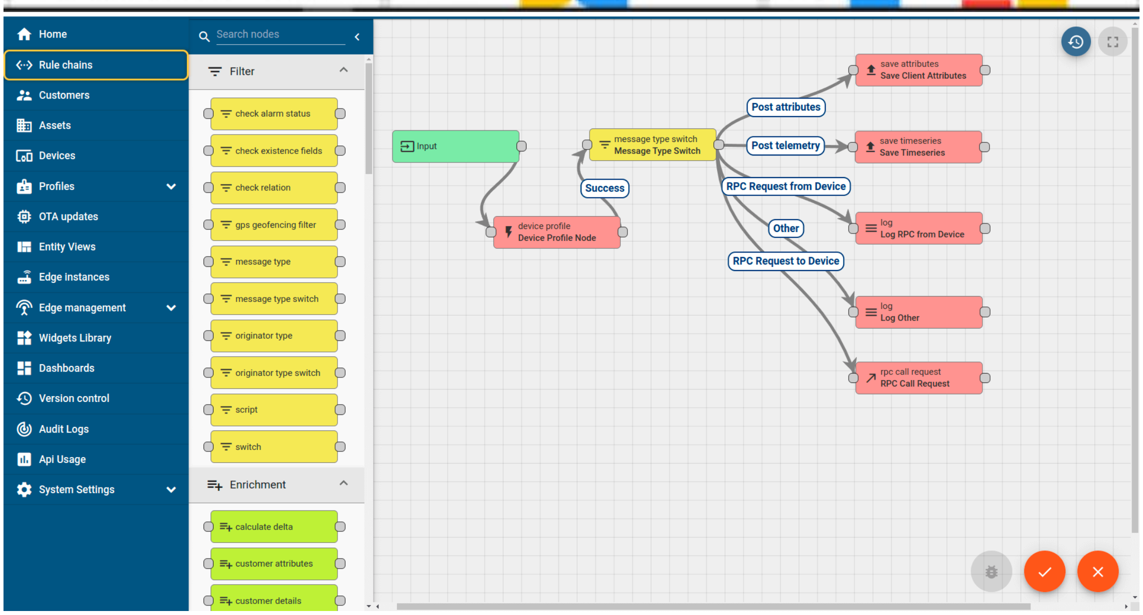
i. UCT IoT Platform (uct Insight)

UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA
- It supports both cloud and on-premises deployments.

It has features to
- Build Your own dashboard
- Analytics and Reporting
- Alert and Notification
- Integration with third party application(Power BI, SAP, ERP)
- Rule Engine




Home
Rule chains
Customers
Assets
Devices
Profiles
OTA updates
Entity Views
Edge instances
Edge management
Widgets Library
Dashboards
Version control
Audit Logs
Api Usage
System Settings
Search nodes
Filter
check alarm status
check existence fields
check relation
gps geofencing filter
message type
message type switch
originator type
originator type switch
script
switch
Enrichment
calculate delta
customer attributes
customer details
Input
message type switch
Message Type Switch
Success
device profile
Device Profile Node
Post attributes
Post telemetry
RPC Request from Device
Other
RPC Request to Device
save attributes
Save Client Attributes
save timeseries
Save Timeseries
log
Log RPC from Device
log
Log Other
rpc call request
RPC Call Request

ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

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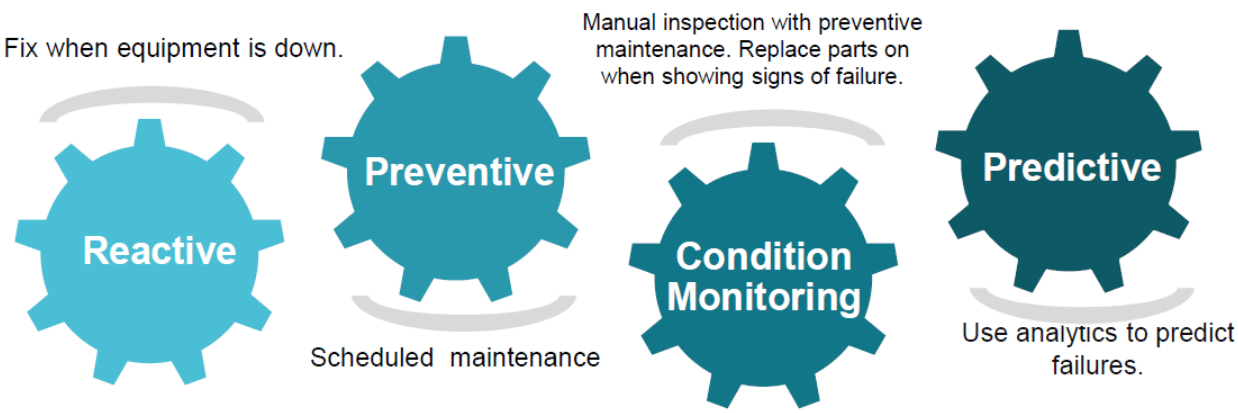
Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i

iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

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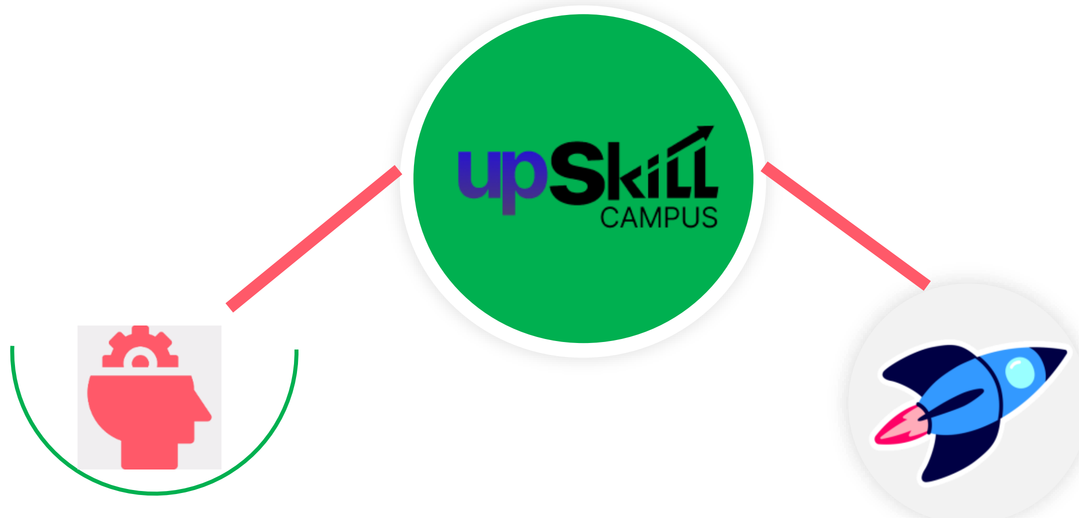
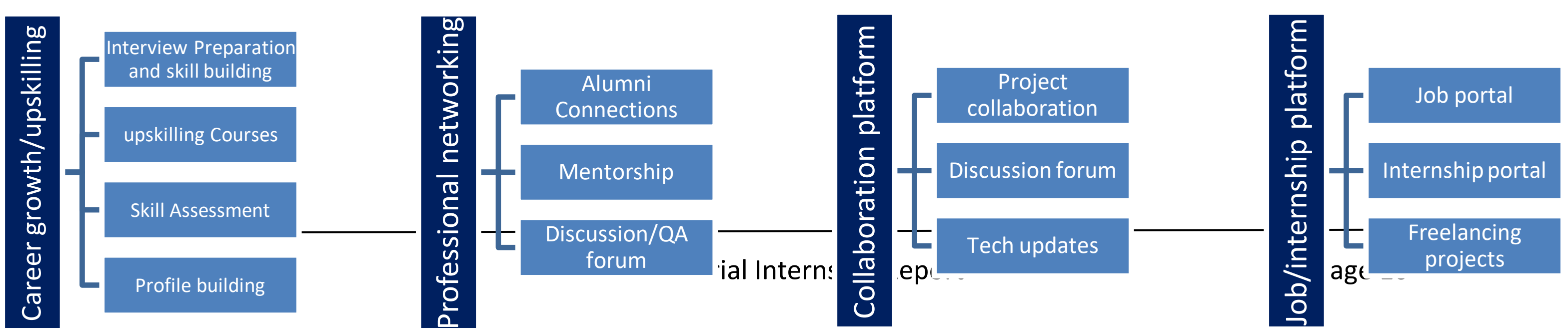


Seeing need of upskilling in self paced manner along-with additional support services e.g. Internship, projects, interaction with Industry experts, Career growth Services

upSkill Campus aiming to upskill 1 million learners in next 5 year

https://www.upskillcampus.com/



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

☛ To gain practical knowledge of Python programming.

☛ To design a Python-Based URL Shortener.

☛ To improve problem-solving skills.

☛ To understand backend development and databases

☛ To enhance debugging and testing skills

2.5 Reference

[1] Python Official Documentation – https://docs.python.org/3/

[2] Flask Documentation – https://flask.palletsprojects.com/

[3] SQLite Documentation – https://www.sqlite.org/

2.6 Glossary

Terms	Acronym
URL	Uniform Resource Locator – the address of a web page.
HTTP	HyperText Transfer Protocol used for communication on the web.
HTTPS	Secure version of HTTP.
FLASK	A lightweight Python web framework used for developing web applications
API	Application Programming Interface used for communication between software components.

3 Problem Statement

3.1 [Python-Based URL Shortener System

In today's digital world, websites often contain lengthy and complex URLs that are difficult to remember, share, and manage. These long URLs are inconvenient when used in emails, social media posts, presentations, or printed documents. They also reduce readability and occupy unnecessary space.

The objective of this project is to develop a **Python-Based URL Shortener System** that converts long URLs into short, unique, and easy-to-share links. Whenever a user enters a valid URL, the system generates a unique short code and stores the mapping between the original URL and the shortened URL in a database. When the shortened URL is accessed, the application automatically redirects the user to the original website.

The proposed system aims to provide a simple, efficient, and reliable solution for managing URLs while improving user convenience. It also demonstrates the practical application of Python programming, backend development, database management, and web technologies.

3.1.1 Objectives of the Project

- Develop a URL shortening application using Python.
- Generate unique and secure short URLs.
- Store original and shortened URLs in a database.
- Redirect users accurately to the original web pages.
- Validate URLs before processing.
- Provide a simple and user-friendly interface.
- Improve understanding of Python backend development and database integration.]

4 Existing and Proposed solution

Existing Solutions

Several URL shortening services are already available on the internet. Some of the most popular platforms include:

- Bitly
- TinyURL
- Rebrandly
- Short.io
- Ow.ly

These platforms allow users to create shortened URLs, monitor click statistics, customize links, and manage multiple URLs.

4.1.1 Limitations of Existing Solutions

Although existing services provide many useful features, they have certain limitations:

- Many advanced features require paid subscriptions.
- Custom short URLs are often available only in premium plans.
- Some services display advertisements before redirecting users.
- Users have limited control over the backend implementation.
- Privacy concerns may arise because URL data is stored on third-party servers.
- Internet connectivity is required to access cloud-based services.

These limitations encourage the development of a simple and customizable URL shortening system that can be implemented locally using Python.

4.2 Proposed Solution

The proposed solution is a **Python-Based URL Shortener System** developed using Python and Flask. The application accepts a long URL from the user, validates it, generates a unique short code, stores the URL mapping in a database, and redirects users whenever the shortened URL is accessed.

4.2.1 Working of the Proposed System

1. The user enters a long URL into the application.
2. The application validates the URL format.
3. A unique short code is generated automatically.
4. The original URL and short code are stored in the database.
5. The shortened URL is displayed to the user.
6. When the shortened URL is opened, the system searches the database.
7. The user is redirected to the original website.

4.3 Advantages of the Proposed System

- Simple and easy to use.
- Fast URL generation.
- Lightweight Python application.
- Efficient database storage.
- Easy maintenance and scalability.
- User-friendly interface.
- Suitable for educational and industrial learning purposes.
- Can be extended with advanced features in the future.

Technologies Used

Technology	Purpose
Python	Core programming language
Flask	Backend web framework
SQLite	Database management
HTML	Web page structure
CSS	User interface styling
JavaScript	Basic client-side interaction
Git	Version control

4.4 Code submission (Github link)

https://github.com/aayushh022/upskillcampus/tree/main/UrlShortner.py

4.5 **Report submission (Github link) :** first make placeholder, copy the link.

https://github.com/aayushh022/upskillcampus/tree/main/UrlShortner_AYUSH_USC_UCT.pdf

5 Proposed Design/ Model

The proposed **Python-Based URL Shortener System** is designed to simplify the process of converting long URLs into short and manageable links. The application is developed using Python and follows a client-server architecture. It accepts a long URL from the user, validates the input, generates a unique short code, stores the URL mapping in a database, and redirects users to the original URL whenever the shortened link is accessed.

The system is designed with modular components to improve maintainability, scalability, and performance. Each module performs a specific task such as user interaction, URL processing, database management, and redirection.

5.1 High Level Diagram (if applicable)

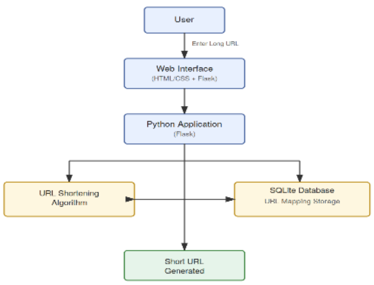
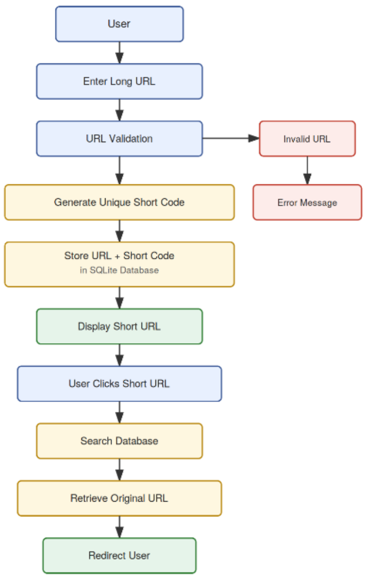


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)



6 Performance Test

This is very important part and defines why this work is meant of Real industries, instead of being just academic project.

Here we need to first find the constraints.

How those constraints were taken care in your design?

What were test results around those constraints?

Constraints can be e.g. memory, MIPS (speed, operations per second), accuracy, durability, power consumption etc.

In case you could not test them, but still you should mention how identified constraints can impact your design, and what are recommendations to handle them.

6.1 Test Plan/ Test Cases

Test Case ID	Test Scenario	Input	Expected Result	Status
TC-01	Generate Short URL	Valid URL	Short URL generated successfully	Pass
TC-02	Invalid URL	Incorrect URL	Display validation error	Pass
TC-03	Empty Input	Blank Field	Display "URL Required" message	Pass
TC-04	URL Redirection	Short URL	Redirect to original website	Pass
TC-05	Duplicate URL	Existing URL	Generate new or existing short URL	Pass
TC-06	Database Storage	Valid URL	Record stored successfully	Pass
TC-07	Invalid Short Code	Random code	Display "URL Not Found"	Pass
TC-08	Multiple Requests	Several URLs	Application performs correctly	Pass

6.2 Test Procedure

The following procedure is proposed to test the application:

6.2.1 Step 1: Environment Setup

- Install Python.
- Install Flask and required libraries.
- Configure the SQLite database.
- Run the application on the local server.

6.2.2 Step 2: Functional Testing

- Enter a valid URL.
- Verify whether a shortened URL is generated.

- Store the generated URL in the database.
- Open the shortened URL.
- Verify that the application redirects to the original website.

6.2.3 Step 3: Validation Testing

- Enter an invalid URL.
- Leave the URL field empty.
- Enter unsupported formats.
- Verify that appropriate error messages are displayed.

6.2.4 Step 4: Database Testing

- Verify that every generated short URL is stored correctly.
- Ensure duplicate short codes are not created.
- Verify that records can be retrieved successfully.

6.2.5 Step 5: Performance Testing

- Generate multiple short URLs.
- Verify response time.
- Check application stability during repeated requests.
- Monitor database performance while handling multiple records.

6.3 Performance Outcome

Based on the proposed testing methodology, the expected performance outcomes are as follows:

Parameter	Expected Outcome
URL Generation	Fast and Accurate
URL Validation	Correct detection of invalid URLs
Database Storage	Successful insertion and retrieval
Redirection	100% accurate for valid URLs

Parameter	Expected Outcome
User Interface	Simple and responsive
Error Handling	Proper validation messages
Scalability	Can handle increasing URL records efficiently

The application is expected to perform efficiently for educational and small-scale deployment purposes. By using Python and SQLite, the system remains lightweight while providing reliable URL shortening and

Testing Summary

The proposed URL Shortener System has been designed with a structured testing approach to ensure reliability and correctness. Functional testing verifies that all major features work as expected, while validation testing ensures the system handles invalid inputs gracefully. Database testing confirms accurate storage and retrieval of URL mappings. Performance testing focuses on response time, scalability, and stability.Although the project is currently in the design and development phase, the prepared testing strategy demonstrates how the application's quality and performance will be evaluated before deploymentredirection functionalit

7 My learnings

The six-week industrial internship provided me with valuable exposure to Python programming and software development practices. Although I was already familiar with basic programming concepts, this internship helped me understand how Python is applied in real-world projects to solve practical problems.

During the internship, I gained knowledge about the complete software development process, including requirement analysis, system design, implementation planning, database management, testing, and documentation. Working on the **Python-Based URL Shortener System** allowed me to understand the logic behind URL shortening, data storage, and web application workflows.

Some of the key technical learnings from this internship include:

- Improved understanding of Python programming fundamentals.
- Learned the basics of backend web development using the Flask framework.
- Understood how HTTP requests and URL redirection work.
- Gained knowledge of SQLite database creation and management.
- Learned how to validate user inputs and handle exceptions.
- Improved debugging and troubleshooting skills.
- Understood the importance of modular programming and clean code practices.
- Learned about project planning and documentation according to industry standards.
- Gained experience in preparing technical reports and maintaining project documentation.

Apart from technical knowledge, I also developed several professional skills such as:

- Analytical and logical thinking.
- Problem-solving ability.
- Time management.
- Technical documentation.
- Professional communication.
- Self-learning through online resources and official documentation.

This internship increased my confidence in Python programming and motivated me to continue learning advanced topics such as REST APIs, cloud deployment, authentication systems, and full-stack web development. Overall, the internship has been an important step toward my career as a software developer.

8 Future work scope

Although the proposed URL Shortener System fulfills the primary objective of generating short URLs and redirecting users to the original web pages, several advanced features can be added in future versions to improve functionality, security, and scalability.

Some possible future enhancements include:

8.1.1 1. User Authentication

Users can create accounts, log in securely, and manage their own shortened URLs.

8.1.2 2. Custom Short URLs

Allow users to choose personalized short links instead of automatically generated codes.

Example:

```
https://short.ly/ayush
```

8.1.3 3. Click Analytics

Provide detailed statistics such as:

- Total Clicks
- Unique Visitors
- Browser Information
- Device Type
- Geographic Location
- Date and Time of Access

8.1.4 4. QR Code Generation

Automatically generate QR codes for every shortened URL to enable quick access through mobile devices.

8.1.5 5. Link Expiration

Allow users to set expiration dates so that shortened URLs become inactive after a specified period.

8.1.6 6. Password-Protected URLs

Provide password security for confidential or private links.

8.1.7 7. Cloud Deployment

Deploy the application on cloud platforms such as:

- Render
- Railway
- PythonAnywhere
- AWS
- Microsoft Azure

This would allow users to access the application online from anywhere.

8.1.8 8. URL Management Dashboard

Develop an admin dashboard where users can:

- View all generated URLs.
- Delete unwanted links.
- Edit URLs.
- Monitor statistics.

8.1.9 9. Enhanced Security

Future versions can include:

- HTTPS enforcement.
- Spam detection.
- Malicious URL filtering.
- Rate limiting.
- CAPTCHA verification.

8.1.10 10. Mobile Application

Develop Android and iOS applications that provide URL shortening services through smartphones.

9. Conclusion

The **Python-Based URL Shortener System** is designed to provide a simple, efficient, and reliable solution for converting lengthy web addresses into shorter, user-friendly URLs. The project demonstrates the practical application of Python programming, database management, and backend web development concepts in solving a real-world problem.

Throughout this internship, I gained valuable exposure to industrial software development practices. I learned how to analyze project requirements, design a system architecture, plan database structures, prepare testing strategies, and document technical work according to professional standards.

Throughout this internship, I gained valuable exposure to industrial software development practices. I learned how to analyze project requirements, design a system architecture, plan database structures, prepare testing strategies, and document technical work according to professional standards.

Although the project is currently at the design and development stage, it provides a strong foundation for implementing a complete URL shortening application in the future. The knowledge gained during this internship has significantly improved my programming skills, logical thinking, and confidence in developing Python applications.

Overall, this internship has been an enriching learning experience that has enhanced both my technical knowledge and professional development. It has also motivated me to continue exploring advanced technologies and pursue a successful career in software development.

References

- Python Software Foundation. *Python Documentation*. https://docs.python.org/3/
- Flask Documentation. https://flask.palletsprojects.com/
- SQLite Documentation. https://www.sqlite.org/docs.html
- Mozilla Developer Network (MDN). https://developer.mozilla.org/
- GitHub Documentation. https://docs.github.com/
- Upskill Campus Learning Resources.
- UniConverge Technologies Internship Materials.